

Leveraging e-transportation in real-time traffic evacuation management

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Abstract

As part of intelligent transportation systems, Internet-connected sensors and cameras have become ubiquitous along roads and highways, enabling many novel e-transportation applications. In this research, we leverage this emerging technology to improve the surface transportation aspect of homeland security, by enhancing its support for evacuation in case of terrorist attacks or other unpredictable disasters. In particular, we extend our existing work on developing a Smart Traffic Evacuation Management System (STEMS), by proposing more efficient evacuation algorithms that dynamically generate evacuation plans for both single and multiple incidents scenarios, based on real-time traffic information obtained from sensor data available through the Web.

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1. Introduction

Recent advancement in sensor technologies has facilitated the development of new e-transportation applications, by providing traffic data with better spatial and temporal resolution, accuracy, precision, repeatability and consistency. As an example, video cameras were first used in traffic management for roadway surveillance based on their ability to transmit *closed-circuit television* (CCTV) imagery to a human operator for interpretation. However, present day technology uses video image processing techniques to automatically analyze the scene of interest and extract real-time information for traffic surveillance and control; this facilitates the development of Internet-based traffic information systems. Currently available traffic-sensing devices provide a wide range of information, from tracking vehicle presence, volume and speed, to detecting traffic density or travel time.

The ubiquitous deployment of sensors in the transportation network, along expressways [1–3] and at intersections of local streets [4–6], is complemented by the increased

accessibility of the collected real-time and historical traffic data over the Web. Several approaches for sensor data dissemination over the Internet have been proposed in the literature. As an example, a browsing environment providing access to real-time and historical traffic data over the Web is the *Freeway Performance Measurement System* (PeMS) [7–9], currently deployed in California. An example of a more generic Web-based architecture for sharing any raw sensor data is the *Internet Sensor Exploration Environment* (iSEE) [10,11], developed at the University of Central Florida. This framework provides a new environment for finding and sharing sensor data and applications over the Internet. Therefore, it encourages a more spread out deployment of sensors and facilitates the development of more and more transportation applications adaptable to real-time traffic conditions. Advances in sensor technologies and Web-based sensor data dissemination have been the two major factors that have motivated us to explore new e-technologies for improving the surface transportation aspect of homeland security.

Intelligent transportation systems (ITS) have attracted great research interest in recent years. However, current ITS applications have been mainly concerned with improving the operational efficiency of the transportation system,

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more exactly, with reducing congestion and expanding personal travel choices. Recently, ITS applications have found a broader function related to homeland security, given the crucial role of the surface transportation system in responding to terrorist attacks or other unpredictable human-caused disasters. This shift, towards advancing homeland security, has occurred in response to the growing concern about the increased threat of terrorism throughout the world. The attacks on the World Trade Center in the United States and the series of coordinated terrorist bombings against the commuter train system of Madrid are only two of the most significant events in the history of escalating terrorist violence over the past decade.

As opposed to other natural disasters that can be anticipated, terrorist attacks are sudden and unexpected. Even if sometimes we do have information about a possible attack, we will not know for sure when, where and how the attack will occur. This lack of information poses great challenges on those responsible for security, in particular on their ability to respond fast whenever necessary with flexibility, efficiency and coordination. Specifically, attacks on the urban transportation system can cause great disruption and alarm, which are the traditional goals of terrorism. Although the surface transportation system is remarkably resilient, it is also open and decentralized, making security responses challenging. Today, Internet-connected sensors are used to both detect and prevent such incidents. Examples of technologies that can detect and head off attacks along rails and roads include surveillance technologies to protect the road infrastructure against tampering, sensors to detect misuse of commercial vehicles carrying hazardous materials, and pattern recognition and analysis capabilities to prevent vehicles carrying hazardous materials from deviating from their pre-planned route [12–14]. Although there are many ITS applications for detecting and preventing terrorist attacks and there have been great successes in identifying and deterring potential threats, it is practically impossible to prevent all acts of terrorism. Therefore, techniques for responding to attacks after they occur are of great interest and critical importance. Our e-transportation application falls in this category, by addressing emergency evacuation in case of terrorist attacks or other unpredictable human-caused disasters. In particular, we explore e-technologies to develop a real-time intelligent traffic evacuation management system for safe and efficient routing of traffic out of a disaster area.

Emergency evacuation is a critical component of life safety. Currently available approaches for addressing evacuation are based on proactive planning [15–17]. This involves developing and testing, in advance, different plans for different scenarios and finding among the available plans the most suitable one to be used whenever an incident occurs. These static predefined schemes are effective only in situations where we know exactly the locations of the incidents, and thus the areas to be evacuated and the typical conditions in those areas (e.g., building evacuation, industrial areas evacuation). In contrast, a traffic network

environment raises more challenges, due to its larger and more dynamic structure, making planning ahead less effective in protecting the evacuees. Currently available tools for proactive evacuation planning and analysis include: the PBS&J model developed by PBS&J Inc. [18], the *Oak Ridge Evacuation Modeling System* (OREMS) developed by Oak Ridge National Laboratory [19] and the *Dynamic Network Evacuation Planning System* (DYNEV) developed by KLD Associate Inc. [20]. Currently, these tools are mostly used in the pre-planning analysis (pre-evacuation stage) and in the post-analysis procedure (post-evacuation stage) [21]. During an evacuation these tools can only assist evacuation staff in analyzing alternative plans.

In contrast to the existing tools, our proposed *Smart Traffic Evacuation Management System* (STEMS) is specifically targeted towards real-time evacuation operation. STEMS can respond rapidly and effectively to human-caused threats and disasters, by automatically constructing evacuation routes given an incident scope and location within the traffic network, and the current traffic conditions. Furthermore, STEMS dynamically controls a set of traffic evacuation signals, placed at intersections, to direct traffic effectively out of the disaster areas according to the generated plans. Since our focus is on terrorist attacks, STEMS is also able to handle multiple coordinated incidents. One specific characteristic of terrorist attacks is that they may occur close in time and proximity to create panic and instability. In this situation not only do we need to direct traffic to avoid a particular incident location, but we also need to revise the evacuation plans currently under execution, to reroute traffic away from all incidents. In this paper we propose evacuation algorithms that enhance STEMS to handle incidents occurring in a short time frame and having overlapping or neighboring evacuation zones. Our problem is more complex than the incident management problem in traffic engineering [22–24], as the latter only handles detouring traffic from a certain damaged or congested street segment, while evacuation implies routing traffic away from many street segments inside the affected areas.

The remainder of this paper is organized as follows. Section 2 is an overview of our initial STEMS framework and Section 3 introduces the extensions that we have proposed to enhance the efficiency and functionality of the initial framework. The simulation results are explained in Section 4, followed by our conclusions in Section 5.

2. Initial STEMS framework

Our initial STEMS [25] was designed to assure the following basic functionality: first, to automatically generate efficient evacuation routes given an incident specified by its intensity and location within the traffic network and the current traffic conditions in the incident area; and secondly, to guide evacuees on the proper routes according to the generated evacuation plan.

Our initial task in implementing the first functionality, the automatic evacuation plan generation, was to develop a model of the street network environment on which to formulate the evacuation problem. We modeled the street network as a graph, with nodes representing intersections, and links corresponding to the street segments between two intersections. Having the street network model, the next problem to address was to define the evacuation problem. Given an incident specified by its location and scope R , we define the *incident node* (IN) as the node closest to the incident location, and the *evacuation zone* (EZ) as the circular area centered at IN with radius R . Then we define the *evacuation exit points* (EEP) as nodes that fall within a certain threshold outside the EZ boundary and are connected to nodes inside the EZ . The evacuation problem becomes directing traffic from all the links inside or intersecting the EZ , to flow away from the incident and exit the affected area via the designated $EEPs$. These concepts are illustrated in Fig. 1. It shows the evacuation routes radiating from the incident node at the center of two circles. The inner circle indicates the boundary of the EZ and the area between the two circles is the threshold area. Nodes that fall within this area and are connected to nodes inside the EZ are chosen as the $EEPs$ and at these points traffic from outside will be blocked from entering the EZ .

Given the area to be evacuated, the second objective was to develop algorithms for constructing the evacuation routes. We have considered two approaches: the *All-Links* approach and the *Fastest-Links* approach. In the first approach, illustrated in Fig. 1, we use the entire street network as evacuation routes. An *evacuation direction* is enforced on each link, such that to guide the traffic out of the EZ . This approach is based on a breadth-first graph traversal; starting from the incident node (IN), we traverse the street network graph such that each link is visited only once. The traversal direction for each link indicates the evacuation direction that traffic will be forced to follow when reaching an intersection, in order to leave the EZ . The *Fastest-Links* approach, illustrated in Fig. 2, is based on multicast routing. The idea is to construct the evacua-



Fig. 1. All-Links approach.



Fig. 2. Fastest-Links approach.

tion routes as a multicast tree having the incident node as the source and the $EEPs$ as receivers.

Due to the nature of our problem the tree construction reduces to overlapping the fastest paths from the incident node to each EEP . The multicast tree would constitute the evacuation routes that all traffic would follow, to leave the EZ via one of the $EEPs$. This tree structure not only provides us with the evacuation routes, but also indicates the desired direction for the traffic flow (from the root to the leaves), allowing us to assign an evacuation direction on each link in the tree. The performance metrics employed for constructing a shortest path is the *aggregated time*, which represents the time required to traverse all links of that path, given that each link is characterized by a certain average speed. Links that are not part of the multicast tree do not have an evacuation direction and they will have bi-directional traffic during evacuation. Note that although initially, all the links have bi-directional traffic, eventually, those that are part of the evacuation routes will only have traffic flowing in the evacuation direction.

Once the evacuation routes are constructed, STEMS has to assure the real-time evacuation management, that is, to direct traffic out of the disaster area according to the generated evacuation plans. To guide the evacuees on the proper routes we use evacuation signals placed at each incoming direction of each intersection. Their design consists of an arrow for each possible outgoing direction. The arrow can take any of the three traditional traffic light colors, namely red, yellow, and green, with the same meaning. At all intersections that are part of the evacuation routes, the evacuation lights will guide incoming traffic to pick one of the outgoing links such that to flow in the evacuation direction. These feasible outgoing directions are displayed as green arrows on the evacuation signals, while unfeasible/undesirable outgoing directions are displayed as red arrows. In our initial design, we rely on drivers' discretion to choose an outgoing direction when more than one is feasible.

To handle conflicting traffic streams competing for the right of way at intersections, we employ a simple timing scheme, which allows equal usage of the intersection among all incoming streams. As for traditional traffic

lights, either pre-timed or actuated signal control strategies, can be employed for controlling our evacuation signals. The *pre-timed control* uses pre-stored signal timing plans and assigns the right of way based on predetermined fixed time durations, for all signal display intervals. This scheme is generally less efficient for controlling intersections that experience frequent changes in demand, as they are insensitive to current traffic volumes. Advances in sensor technologies have enabled a more efficient signal control strategy that overcomes the limitations of the pre-timed control, by adapting to real-time traffic flow. This *actuated control* strategy requires sensors to provide information about vehicles presence to a traffic signal controller. Different sensor technologies can be used to detect vehicle presence, inductive loop detectors embedded in the pavement being the most frequently used ones. Multiple-lane and multiple-detection-zone sensing devices such as CCTVs with video image processors, or ultrasonic sensors can also be used in conjunction with actuated control. The availability of such a wide range of sensing devices makes real-time traffic adaptive control increasingly attractive and enables the development of more efficient algorithms for traffic applications.

3. Framework extensions

In this section we present the two major extensions that enhance the functionality and efficiency of our initial framework [25]. The first extension is an improved algorithm for the evacuation routes construction, which combines the advantages of the *All-Links* and *Fastest-Links* approaches. The second extension improves the functionality of our evacuation management system with the ability to handle multiple incidents occurring in close proximity in a short time frame.

3.1. Hybrid evacuation routes construction algorithm

The main disadvantages of the two *evacuation routes* construction approaches, presented in Section 2, are as follows. The *Fastest-Links* approach may create congestion, by concentrating traffic on the evacuation tree and leaving the other street segments unutilized; on the other hand, the *All-Links* approach, although utilizes the entire street network, does not take into account the different traffic densities of the links. To overcome these drawbacks we propose a *Hybrid* approach which combines the advantages of the previous two techniques as follows: first, it will exploit traffic information as in the *Fastest-Links* approach; and secondly, it will utilize more street segments during evacuation, as in the *All-Links* approach, in order to reduce the congestion that occurs when traffic is concentrated on the evacuation routes.

This *Hybrid* approach, illustrated in Fig. 3, is based on *shortest-path tree* construction. Instead of constructing a tree containing the fastest routes from the incident node to each *EEP* (as for the *Fastest-Links* approach) the *Hybrid*



Fig. 3. Hybrid approach.

approach constructs the tree containing the fastest paths from the incident node *IN* to all the other nodes in the evacuation zone. This *shortest-path tree* would constitute the evacuation routes that all traffic would follow, to leave the *EZ* via one of the *EEPs*. As for the *Fastest-Links* approach, this tree structure will indicate the desired direction for the traffic flow (from the root to the leaves), allowing us to assign an evacuation direction on each link in the tree. Links that are not part of the *shortest-path tree* will have bi-directional traffic during evacuation. The performance metrics employed for constructing the *shortest-path tree* is the *aggregated time*, as defined for the *Fastest-Links* approach in Section 2. The advantage of the *Hybrid* approach is that the evacuation routes will utilize a higher percentage of the street network than the *Fastest-Links* approach, therefore reducing congestion.

In the *Hybrid* approach all nodes in the evacuation zone will be part of the evacuation routes. Therefore, at all nodes evacuation signals will guide incoming traffic to pick one of the outgoing links such that to flow in the evacuation direction. We note that traffic flowing opposite to the evacuation direction may occur only in the early stage of the evacuation. Eventually, all the traffic will be flowing only through the *shortest-path tree* and only in the evacuation direction. Since each node in this tree can have only one parent node, there is only one incoming traffic flow at each intersection in the tree. In this case, if actuated evacuation signals are used, they will detect the absence of traffic from all the other directions, and maintain the right of way to the sole approaching traffic flow. Effectively, timing schemes are no longer necessary in this stage and all traffic flows freely along the *shortest-path tree* towards the *EEPs*. This greatly improves evacuation efficiency.

3.2. Handling multiple incidents

For handling multiple incidents targeting a specific area in a short time frame, our evacuation algorithms have to address two additional challenges: first, to be able to construct evacuation routes for incidents with overlapping

evacuation zones, such that to direct traffic away from all incidents; and secondly, to prohibit congestion in the areas around the evacuation zones from impeding the evacuation execution. This second problem occurs in areas that are in close proximity with more than one incident, and it may drastically affect the evacuation time.

3.2.1. Handling overlapping evacuation zones

As in our previous work [26], we store the street network and the related traffic data in two relational tables as follows:

- *Nodes* table: Each record represents an intersection characterized by the *NodeID*, and the *X* and *Y* coordinates of the location.
- *Links* table: Each record represents a road segment characterized by its two end points (*NodeID1* and *NodeID2*) the average speed values in each direction (*Speed₁₂* and *Speed₂₁*) and an *EvacuationDirection*, which is initially set to “0” and during evacuation is revised to “1” or “2”, to indicate the desired direction for the traffic flow on that street segment.

The above data structure contains two types of data: static and dynamic. Static data consists of the topology of the street network, that is, the street segments and the intersections and their geographical location. While this data does not change frequently in time, the average speed values for individual street segments are much more dynamic, varying in time according to several parameters (e.g., the time of the day, weather conditions, etc.). Average (or space mean) speed is defined as the arithmetic mean of the speed of vehicles occupying a given street segment. Some of the widely used sensing devices for measuring speed include: inductive loops, pulsed ultrasound, single detection zone passive infrared and magnetometers. These sensors are used in pairs and the speed is calculated as the ratio of the distance between the sensors and the time it takes a vehicle to traverse that distance. Another way to measure speed is with multi-detection zone sensors such as a video image processor, multi-zone passive infrared, microwave radars, or acoustic arrays. For example, a video image processing technique can measure speed based on vehicle movement across a calibrated distance in the field of view. The wide dissemination of such traffic data over the Internet has enabled us to enhance e-transportation by developing more advanced evacuation algorithms that could take advantage of accurate real-time traffic information.

Using the *EvacuationDirection* column for capturing evacuation routes, the *Links* table will maintain a global evacuation plan for all incidents, and update it with new plans whenever new incidents occur. The procedure is as follows. When an incident occurs, the corresponding *EZ* is computed as in a single-incident scenario (presented in the previous section). The *EZ* computation algorithm identifies the subgraph of the street network graph, affected by

the incident. The leaf nodes, the non-leaf nodes and the links of this subgraph are stored in three temporary tables: *EEPNodes*, *EZNodes* and *EZLinks*, respectively. This subgraph is used as input by the evacuation routes construction algorithm (the All-Links, the Fastest-Links or the Hybrid Approach) to generate evacuation routes, which are then captured in the *EZLinks* table in the following way: for each directed link *L* part of the evacuation routes, the tuple (*NodeID1*, *NodeID2*, *EvacuationDirection*) is identified in the *EZLinks* table, and the value of the *EvacuationDirection* is set to “1” if the direction of *L* is from *NodeID1* to *NodeID2*, and to “2” otherwise.

In case of multiple incidents, the evacuation routes of a new incident may overlap with existing plans generated by previous incidents and may result in conflicting evacuation directions on some of the street segments. These conflicts occur in areas where the *EZs* of two or more incidents overlap. However, not all street segments in the overlapping areas will experience conflicts. Given several overlapping incidents, if their corresponding evacuation plans assign the same evacuation direction on a street segment, then traffic following that direction will flow away from all incidents. This is because an evacuation plan is constructed such that to direct traffic away from the given incident. Conversely, if a street segment is assigned different evacuation directions, neither of those directions will direct traffic away from all incidents. As evacuation should not direct traffic towards any incident, but away from all of them, we do not enforce an evacuation direction on street segments that experience conflicts. In other words, we remove those street segments from the evacuation plans of both current and previous incidents. We illustrate this with the example in Fig. 4. The initial evacuation plan for the smaller incident (shown in Fig. 1) changes when the larger incident occurs; the dotted-line street segments (inside or intersecting the overlapping area) are the ones assigned conflicting evacuation directions and are removed from the evacuation routes of both incidents. This change assures that whenever a new incident occurs, the previous evacuation plans, currently under execution, are updated such that they do not direct traffic towards the new incident.

To differentiate between street segments that were never assigned an evacuation direction (because they were not part of any evacuation plan, so their *EvacuationDirection* is “0”) and the ones that were assigned a direction that was later removed due to conflicts, we set the *EvacuationDirection* of the later ones to “3”. The evacuation routes for the currently handled incident, captured in the *EZLinks* table, are overlaid on the global evacuation plan resulting from previous incidents (stored in the *Linkstable*) according to the following rule: if the currently computed evacuation direction for a street segment (from the *EZLinks* table) conflicts with the one already assigned as a result of previous incidents (captured by the *Links* table) the *EvacuationDirection* in the *Links* table is set to “3”; otherwise it is set to the newly computed value (from the

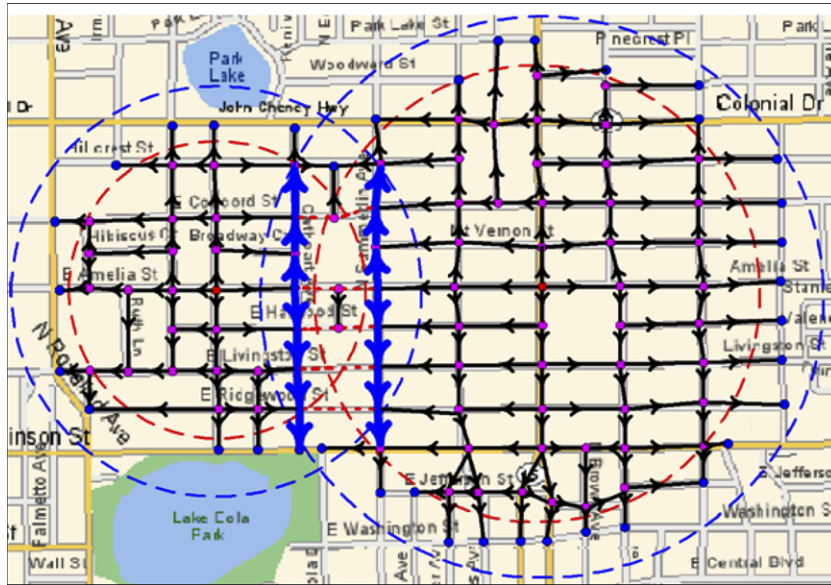


Fig. 4. All-Links evacuation for two incidents with overlapping EZs.

EZLinks table). We note that once the *EvacuationDirection* of a street segment becomes “3” it will not change anymore, as future assignments will always result in conflicts.

To guide traffic according to the generated global evacuation plan, we control the evacuation signals at the intersections that are part of the evacuation routes, by imposing some restrictions. More exactly, incoming traffic is only allowed to pick one of the outgoing links such that to flow in the evacuation direction. If this choice is not available, traffic is guided to pick one of the links with no evacuation direction (such that not to flow opposite to an evacuation direction). If such a choice is not available either, then traffic is allowed to pick any of the outgoing links and will be guided in the evacuation direction when it reaches the next intersection. At nodes that are not on the evacuation routes traffic proceeds as in normal conditions, that is, drivers may pick any outgoing direction.

The above restrictions assure that traffic will not leave an evacuation route until it ends. Fig. 4 shows that all the traffic flowing towards the overlapping area will be stopped at the thick-line street segments and directed along them, away from both incidents. Similarly, traffic inside the overlapping area will be blocked from flowing towards the incidents by being directed along the thick-line streets and thus away from both incidents.

3.2.2. Congestion avoidance using secondary zone evacuation

As stated previously, another problem associated with multiple incidents, is when the *evacuation zones* (EZs) of the incidents are not necessarily overlapping, but they are close enough that the area between them becomes congested and impedes the evacuation execution. To handle this problem we define a *secondary zone* (SZ) around the existing EZs and evacuate traffic from this area in order to facilitate the evacuation of the traffic in danger from inside the evac-

uation zones. An SZ is defined as follows. Whenever an incident occurs we first check whether its evacuation area is in close proximity with any of the previous incidents. If the difference between the distance connecting two incidents and the sum of their radii is higher than a certain threshold value then we consider the incidents independent; otherwise, they are considered in close proximity (even though they are not necessarily overlapping). If two or more incidents are in close proximity then the *minimum bounding rectangle* (MBR) enclosing their EZs is first computed and then the secondary zone is defined as the disk having the same center as the MBR and diameter equal to the larger side of the MBR. In Fig. 5, the black disks represent the EZs and the gray one represents the secondary zone. The SZ captures the areas around the EZs that are susceptible to congestion. The more critical street segments are the ones in the dark grey area, as they receive traffic from more than one incident.

To avoid congestion and facilitate the evacuation of the traffic in danger from inside the EZs we also evacuate the traffic from the SZ. Any of the three techniques (*All-Links*, *Fastest-Links* or *Hybrid*) can be applied for SZ evacuation

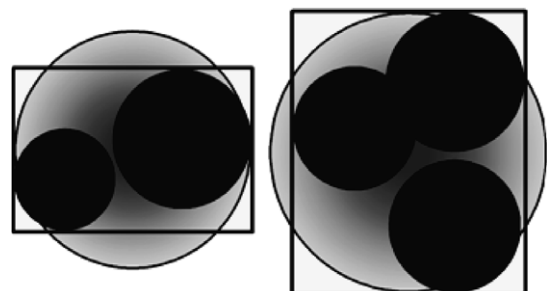


Fig. 5. Secondary zone construction.

assuming that we have a virtual incident in the center of the SZ. We note here that a naïve approach would be to simply replace several incidents with their corresponding virtual incident and then evacuate traffic by constructing the evacuation routes for the virtual incident. However, in this situation, by guiding traffic away from the virtual incident we will be directing it towards the actual incidents. This undesirable effect does not allow us to ignore the EZs of the actual incidents. Our approach is to assign an evacuation direction only on streets that are not part of any EZ (the gray areas in Fig. 5), when constructing the evacuation routes for the SZ. Thus, streets that belong to a certain EZ will maintain their previously assigned evacuation directions, assuring that traffic in each EZ will flow away from its corresponding incident. This approach decreases the evacuation times significantly. We note that there are other ways we can construct the secondary zone, for example, instead of using the larger side of the MBR, we can use the smaller one (if we want a smaller SZ) or the diagonal (if we want a larger SZ). We can also investigate different shapes of the SZ in order to better capture the areas susceptible to congestion, but we plan to address these issues in our future work.

4. Simulation study

Due to the nature of our work, modeling and simulation methods are essential elements in the design and evaluation of STEMS. The TSIS (*Transportation System Integrated Software* [27]) traffic simulator, built as a shell around the well-known microscopic simulator Corsim [28], was used for our performance study. In this section we present the simulation environment and then discuss the results of the simulations that we have carried out for assessing the effectiveness and efficiency of the STEMS algorithms in handling multiple incidents.

4.1. Simulation environment

For our performance studies we use the urban street network from the downtown Orlando in Florida. We present two multiple incidents scenarios: (1) three incidents having overlapping evacuation zones, with 79, 78 and 73 intersections, and 109, 110 and 105 road segments, respectively (illustrated in Fig. 6); and (2) three incidents occurring in close proximity (but without overlapping evacuation zones), having 76, 68, 75 intersections and 105, 95, 107, street segments, respectively (illustrated in Fig. 7). The first scenario is used for investigating the effectiveness of our algorithms in handling conflicting flows in the overlapping areas, and the second scenario is used for assessing the role and efficiency of the SZ evacuation.

4.2. Simulation results

The performance metric used to compare the *All-Links*, *Fastest-Links* and *Hybrid* approach (with and without SZ



Fig. 6. Scenario 1.

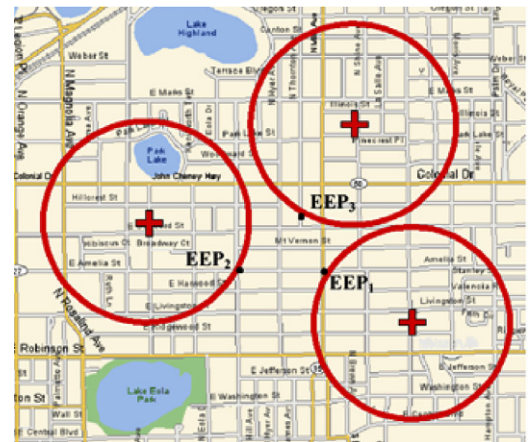


Fig. 7. Scenario 2.

evacuation) is *Evacuation Effectiveness (EE)*. As defined in [25], *EE* measures how fast is traffic evacuated. It can be plotted as the number of evacuees inside an EZ over time. We first investigate the overlapping incidents scenario to assess the effectiveness of our algorithms in handling conflicting flows in the overlapping areas of the evacuation zones.

We present the *EE* for each individual evacuation zone and also the global *EE* for all three EZs in Fig. 8. The three peaks in the global *EE* correspond to the occurrence times of the three incidents, each increasing the evacuation zone and thus the number of evacuees. The first incident occurs at time 0, the second incident occurs 80 seconds later, and the third incident occurs 120 seconds after the second. We notice that in all graphs from Fig. 8 the number of evacuees in each EZ is monotonically decreasing, meaning that later incidents do not cause more traffic to enter the EZ of previous incidents.

Fig. 8 also shows that by evacuating the SZ, the evacuation times are significantly reduced. This is emphasized in Fig. 9. For the *All-Links* approach the evacuation is almost completed after 32 min if the SZ evacuation is used, while

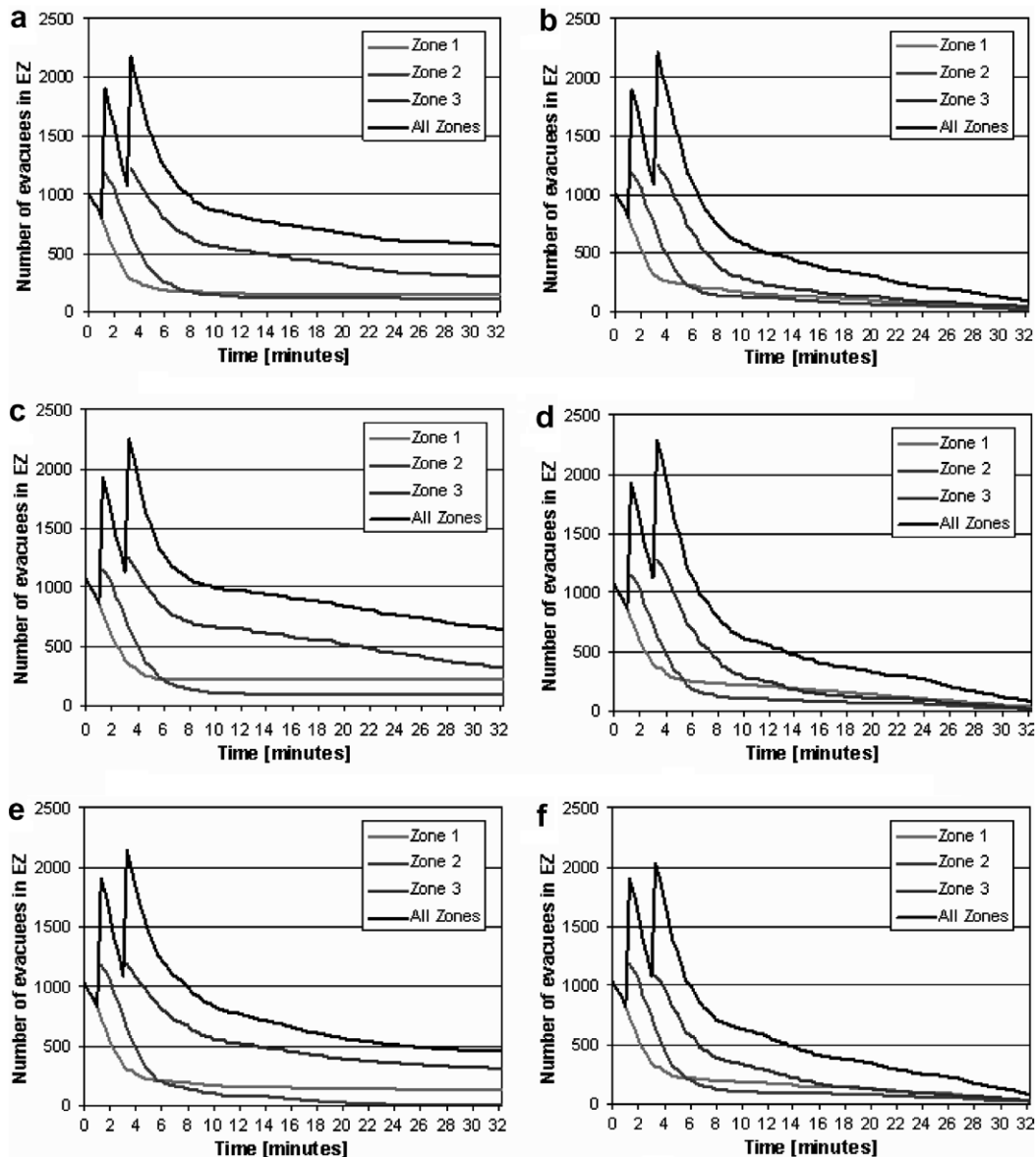


Fig. 8. Evacuation effectiveness for Scenario 1: (a) All-Links without SZ, (b) All-Links with SZ, (c) Fastest-Links without SZ, (d) Fastest-Links with SZ, (e) Hybrid approach without SZ and (f) Hybrid approach with SZ.

there are approximately 500 more cars to be evacuated when the *SZ* is not used. Similarly, for the *Fastest-Links* approach, the difference is about 600 cars, and for the *Hybrid* approach it is approximately 250 cars. It is worth mentioning that due to the nature of evacuation, performance improvement has a different meaning than in other computer related applications. Since we are dealing with people's safety and human lives, even a small increase in performance has an extremely significant value.

To further assess the role and effectiveness of the secondary evacuation zone in multiple incidents evacuation we perform simulations on our second scenario (Fig. 7). We summarize the performance of *All-Links*, *Fastest-Links* and *Hybrid* approach, for the non-overlapping incidents scenario, with and without *SZ* evacuation, in Fig. 10. We observe the same behavior as for our first scenario: without

evacuating the *SZ*, all three approaches result in evacuation times significantly higher than when the *SZ* is evacuated. In the *All-Links* approach, the three *EZ* are empty after 32 min if *SZ* evacuation is enforced, while there are about 400 cars still left, if the *SZ* is not evacuated. In comparing the two scenarios, the performance improvement achieved by the *Fastest-Links* approach is higher than that of the *All-Links* and *Hybrid* approach in the first scenario, while the *All-Links* approach exhibits the highest performance gain in our second scenario.

We also compare the performance of *All-Links*, *Fastest-Links* and the *Hybrid* approach in Figs. 11 and 12 for Scenarios 1 and 2. The graphs show that the three techniques perform almost similar in Scenario 1 when *SZ* is evacuated and the *Hybrid* approach behaves the best when *SZ* evacuation is not used. In Scenario 2 (Fig. 12), the *Hybrid*

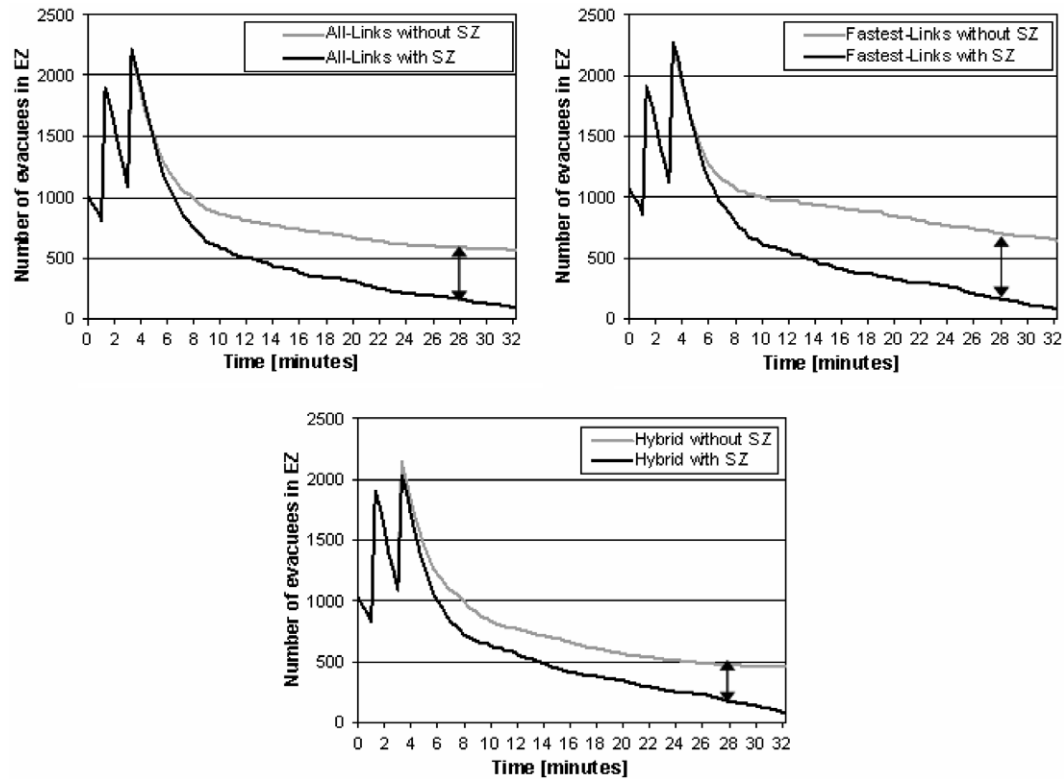


Fig. 9. SZ evacuation effectiveness comparison for scenario 1.

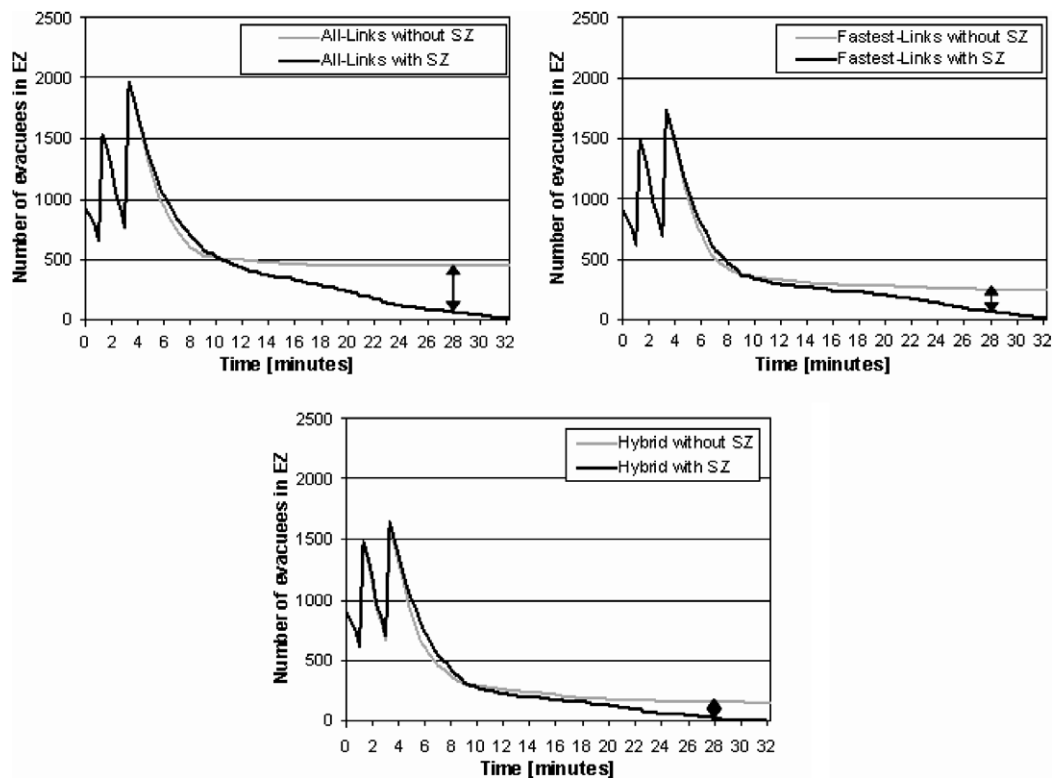


Fig. 10. SZ evacuation effectiveness comparison for scenario 2.

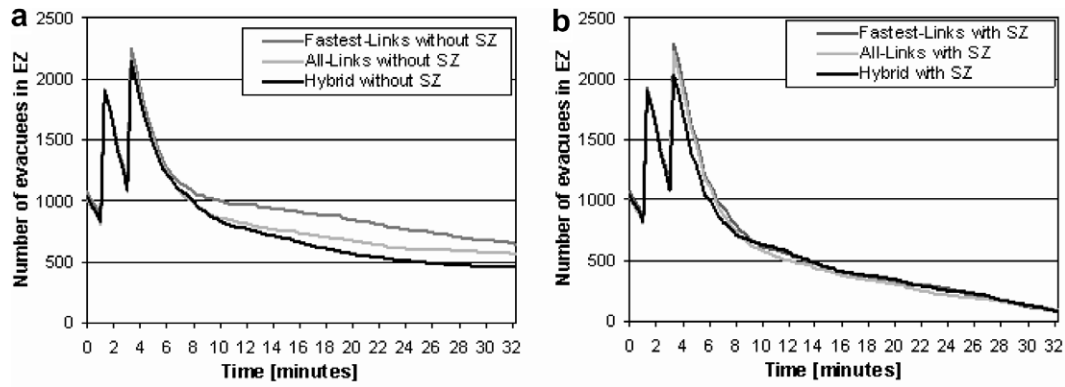


Fig. 11. Scenario 1: All-Links vs. Fastest-Links vs. Hybrid approach: (a) without SZ and (b) with SZ.

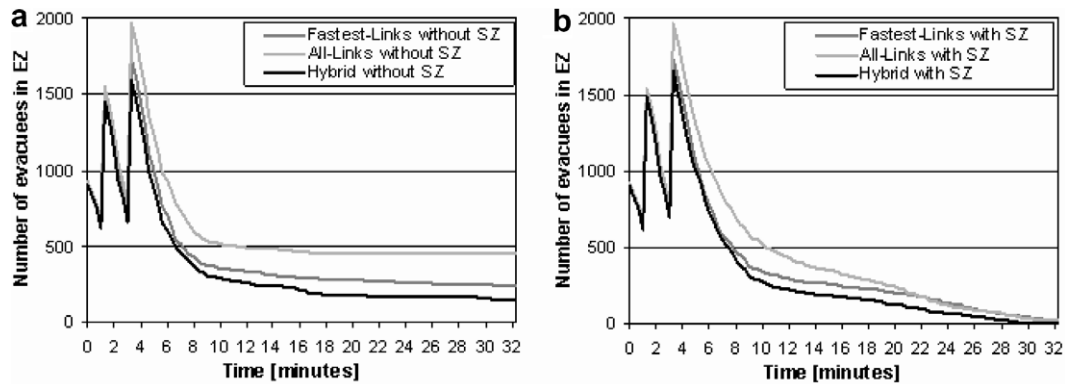


Fig. 12. Scenario 2: All-Links vs. Fastest-Links vs. Hybrid approach: (a) without SZ and (b) with SZ.

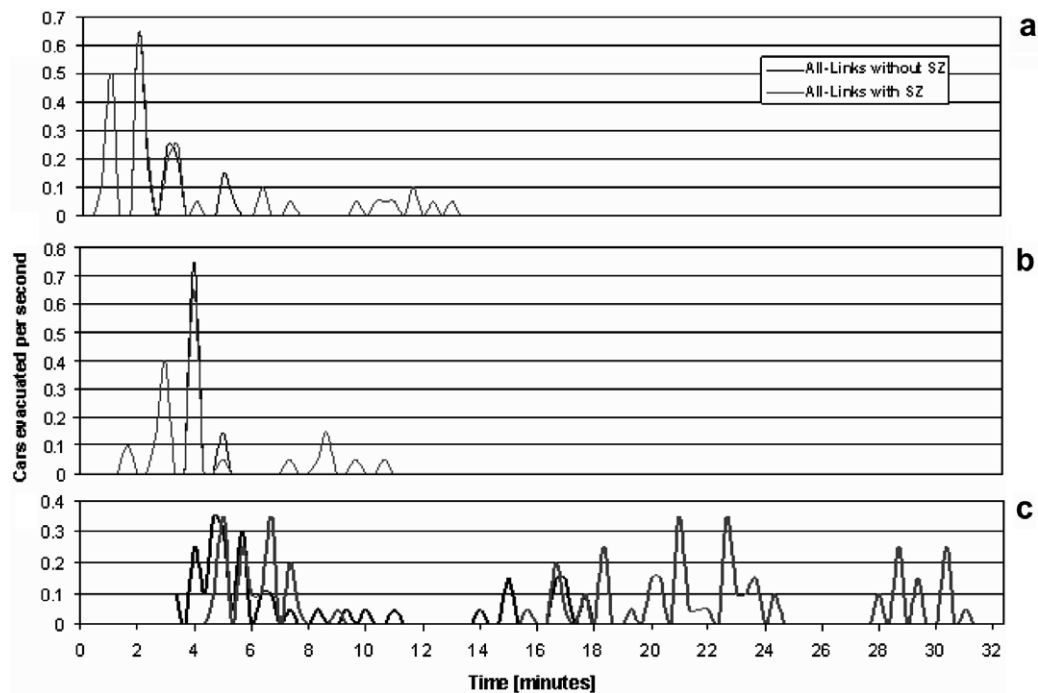


Fig. 13. Evacuation throughput at (a) EEP_1 , (b) EEP_2 , and (c) EEP_3 .

approach performs better than both *All-Links* and *Fastest-Links*, especially when *SZ* evacuation is not applied. Thus, we can conclude that, as expected, the *Hybrid* approach always outperforms both *All-Links* and *Fastest-Links* techniques.

To have better insight into the role of *SZ* evacuation we investigated the throughput of one *EEP* from each evacuation zone. The chosen *EEPs* are shown in Fig. 7. The evacuation throughput at these points, measured as vehicles/second, is shown in Fig. 13 (for the *All-Links* approach). We observe that if the *SZ* is not evacuated the throughputs at the EEP_1 and EEP_2 become close to 0 after 6 min, and at the EEP_3 after 18 min. When *SZ* evacuation is applied the throughput at each point continues to have high values until evacuation of the corresponding zone is completed.

5. Conclusions

Continuous advances in sensor technologies have enabled the development of more efficient e-transportation applications by providing more and more information about real-time traffic. In this paper we explore the advantages of employing Internet-connected sensors to enhance e-transportation with the ability to respond to terrorist threats and other human-caused disasters. The surface transportation system is of critical importance in responding rapidly and effectively to various human-caused disasters. The significance of our work consists of enhancing the surface transportation aspects of homeland security by leveraging e-technologies in the street network to develop a *Smart Traffic Evacuation Management System* (STEMS). The evacuation algorithms employed by STEMS are effective, generating efficient evacuation plans for both single and multiple incidents scenarios. To the best of our knowledge, STEMS is the first system that dynamically generates evacuation plans given incidents locations, their scopes, and the real-time traffic conditions obtained through various sensors over the Internet, and subsequently, automatically controls traffic lights to direct evacuation traffic away from all incidents in a safe manner.

One of the limitations of this research is that in our simulation studies we assumed that the drivers behave in a safe manner, obeying traffic rules without making mistakes or misjudgments and following the evacuation signals directions. This, however, is true only in an ideal world and it might be inadequate in emergency situations when panic and fear might push drivers to lose their self-control and behave in unpredictable ways. Also, the inherent agitation in a crisis situation might distract drivers, leading to crash occurrences and increased tension. Future research should explore developing driver behavior models that incorporate the panic and tension inherent in crisis situations, and investigate the relative behavior of our proposed evacuation algorithms when applied in more realistic traffic environments that better emulate the real driver behavior.

Due to the critical nature of the problem that we address, the algorithms for real-time traffic evacuation must generate the evacuation plans quickly, to ensure timely and effective response to emergencies. Given the complex nature of the traffic environment, optimal solutions to the evacuation problem are too computationally expensive. Therefore, future research efforts should focus on investigating techniques to improve the efficiency of the algorithms employed by STEMS while maintaining their computational complexity low to assure they can be used in a real-time evacuation operation. As an example, we can investigate deploying different secondary evacuation zone shapes and sizes that better capture the areas around the evacuation zones that are prone to congestion.

In addition to guiding traffic out of the evacuation zone as fast and effective as possible, future research should also focus on diverting traffic in the vicinity of the evacuation zone by providing detour routes that traffic can follow to go around the disaster areas. This detouring must be done with great care to minimize its effect on the evacuating traffic.

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References

- [1] Z. Yin, F. Yang, H. Liu, B. Ran, Using image sensors to measure real-time traffic flow parameters, in: TRB 83rd Annual Meeting, Washington DC, 2004, Preprint CD-ROM.
- [2] X.D. Yu, L.Y. Duan, Q. Tian, Highway traffic information extraction from Skycam MPEG video, in: 5th IEEE Int'l. Conf. on Intelligent Transportation Systems, Singapore, 2002, pp. 37–42.
- [3] Z. Yin, L. Cao, Traffic parameters detection based on video virtual-loop technologies, in: 7th Int'l. Conf. on Application of Advanced Technology in Transportation, Boston, MA, 2002, pp. 885–893.
- [4] M. Fathy, M.Y. Siyal, A window-based edge detection technique for measuring road traffic parameters in real-time, *Real-Time Imaging* 1 (4) (1995) 297–305.
- [5] M. Fathy, M.Y. Siyal, Processing approach to measure traffic queue parameters, vision, *Image and Signal Processing* 142 (5) (1995) 297–303.
- [6] J. Soh, B.T. Chun, M. Wang, Analysis of road sequences for vehicle counting, in: Int'l. Conf. on Systems, Man and Cybernetics, Vancouver, Canada, 1995, pp. 679–683.
- [7] J. Kwon, C. Chen, P. Varaiya, Statistical methods for detecting spatial configuration errors in traffic surveillance sensors, *Transportation Research Record* 1870 (2005) 124–132.
- [8] T. Choe, A. Skabardonis, P. Varaiya, Freeway performance measurement systems (PeMS): an operational analysis tool, *Transportation Research Record* 1811 (2002) 67–75.
- [9] C.K. Ng, C. Chen, Wireless content delivery and user profiling, in: IEEE 4th Int'l. Conf. on Intelligent Transportation Systems, Oakland, CA, 2001, pp. 973–975.
- [10] R. Peng, K.A. Hua, G.L. Hamza-Lup, A Web services environment for internet-scale sensor computing, in: IEEE Int'l. Conf. on Services Computing, Shanghai, China, 2004, pp. 101–108.

- [11] K.A. Hua, R. Peng, G.L. Hamza-Lup, Dissemination of sensor data over the Internet, in: *Int'l. Conf. on Embedded and Ubiquitous Computing*, Aizu, Japan, 2004, pp. 745–754.
- [12] H. Veeraraghavan, O. Masoud, N.P. Papanikolopoulos, Computer vision algorithms for intersection monitoring, *Transactions on Intelligent Transportation Systems* 4 (2) (2003) 78–89.
- [13] B. Coifman, D. Beymer, P. McLauchlan, J. Malik, A real-time computer vision system for vehicle tracking and traffic surveillance, *Transportation Research: Part C* 6 (4) (1998) 271–288.
- [14] M. Abkowitz, P.-M. Cheng, M. Lepofsky, Use of geographic information systems in managing hazardous materials shipments, *Transportation Research Record* 1261 (1990) 35–43.
- [15] T.M. Kiosko, R.L. Francis, C.R. Nobel, *Evacnet4 User's Guide*, University of Florida, 1998.
- [16] B. Hoppe, E. Tardos, Polynomial time algorithms for some evacuation problems, in: *5th Annual ACM-SIAM Symp. on Discrete Algorithms*, Arlington, VA, 1994, pp. 433–441.
- [17] Q. Lu, Y. Huang, S. Shekhar, Evacuation planning: a capacity constrained routing approach, in: *First NSF/NIJ Symp. on Intelligence and Security Informatics*, Tucson, AZ, 2003, pp. 111–125.
- [18] PBS&J, Southeast United States hurricane evacuation traffic study: evacuation travel demand forecasting system, Technical Memorandum 2, Post, Buckley, Schuh and Jernigan Inc., 2000.
- [19] O. Franzese, *OREMS 2.6 User's Guide*, Oak Ridge National Laboratory, 2003.
- [20] KLD, *Evacuation Planning Services*, KLD Associates Inc., 2001.
- [21] E. Chang, *Traffic Simulation for Effective Emergency Evacuation*, Oak Ridge National Laboratory, 2003.
- [22] L.D. Aronson, R.P.J. van der Krogt, J. Zutt, Incident management in transport planning, in: *7th Congress on Transport, Infrastructure and Logistics*, Rotterdam, Netherlands, 2002.
- [23] A. Stentz, Optimal and efficient path planning for partially-known environments, in: *IEEE Int'l. Conf. on Robotics and Automation*, San Diego, CA, 1994, pp. 3310–3317.
- [24] A. Stentz, The focussed d* algorithm for real-time replanning, in: *14th Int'l. Joint Conf. on Artificial Intelligence*, Montreal, Canada, 1995, pp. 1652–1659.
- [25] G.L. Hamza-Lup, K.A. Hua, R. Peng, Applying e-transportation for traffic evacuation management under human-caused threats, in: *IEEE Int'l. Conf. on e-Technology, e-Commerce and e-Service*, Hong Kong, China, 2005, pp. 666–673.
- [26] G.L. Hamza-Lup, K.A. Hua, M. Le, R. Peng, Enhancing intelligent transportation systems to improve and support homeland security, in: *7th IEEE Int'l. Conf. on Intelligent Transportation Systems (ITSC'04)*, Washington, DC, 2004, pp. 250–255.
- [27] L. Owen, Y. Zhang, L. Rao, G. McHale, Traffic flow simulation using Corsim, in: *2000 Winter Simulation Conf.*, Orlando, FL, 2000, pp. 1143–1147.
- [28] S.B. Caldwell, *CORSIM 3.0 User and Technical Documentation*, Cornell University, 1997.